The expectation of a function $g(x)$

If $g(x)$ is any function of a continuous random variable X with a p.d.f$f(x)$, then the expected value of $g(x) = \int g(x) \times f(x)dx \; for \; all \; x$

Example

A random variable x has the following p.d.f

$$f(x) = \begin{cases} \frac{1}{20}(x+3) \;\; 0 \leq x \leq 4 \\ 0 \;\; elsewhere \end{cases}$$

Find

a) $E(x)$
b) $E(2x+5)$
c) $E(x^2)$
d) $E(x^2+2x+5)$

Solution

a) $E(x)$

$E(x) = \int x \times f(x)\, dx$

$= \frac{1}{20}\int_0^4 (x+3)x\, dx$

$= \frac{1}{20}\int_0^4 x^2 + 3x\, dx$

$= \frac{1}{20}[\frac{x^3}{3} + \frac{3x^2}{2}]_0^4$

$= \frac{1}{20}(45\frac{1}{3})$

$mean = \frac{34}{15} = 2.2667$

b) $E(2x+5) = E(2x) + E(5)$

$= 2E(x) + E(5)$

$= 2 \times \frac{34}{15} + 5 = 9.53$

c) $E(x^2)$

$E(x^2) = \int_0^4 x^2 \times f(x)dx$

$= \frac{1}{20}\int_0^4 x^2\ (x+3)dx$

$= \frac{1}{20}\int_0^4 x^3 + 3x^2\ dx$

$\frac{1}{20}[\frac{x^4}{4} + \frac{3x^3}{3}]_0^4 = \frac{1}{20}[\frac{x^4}{4} + x^3]_0^4$

$\frac{1}{20}(64+64) = \frac{128}{20} = 6.4$

d) $E(x^2 + 2x + 5)$

$E(x^2) + 2Ex + E(5)$

$6.4 + 2 \times \frac{34}{15} + 5 = 15.93$

Variance

$var(x) = E(x^2) - [E(x)]^2$

$\int x^2 \times fx\ dx - (\int x \times fx\ dx)^2$ For all x

Example

Given $f(x) = \frac{1}{20}(x+3)$ $0 \leq x \leq 4$ obtain variance x

$var(x) = E(x^2) - [E(x)]^2$

$E(x^2) = \int_0^4 x^2 \times f(x)dx$

$= \frac{1}{20}\int_0^4 x^2\ (x+3)dx$

$= [\frac{x^4}{4} + x^3]_0^4 = 128$

$= \frac{128}{20} = 6.4$

$E(x) = \frac{34}{15}$

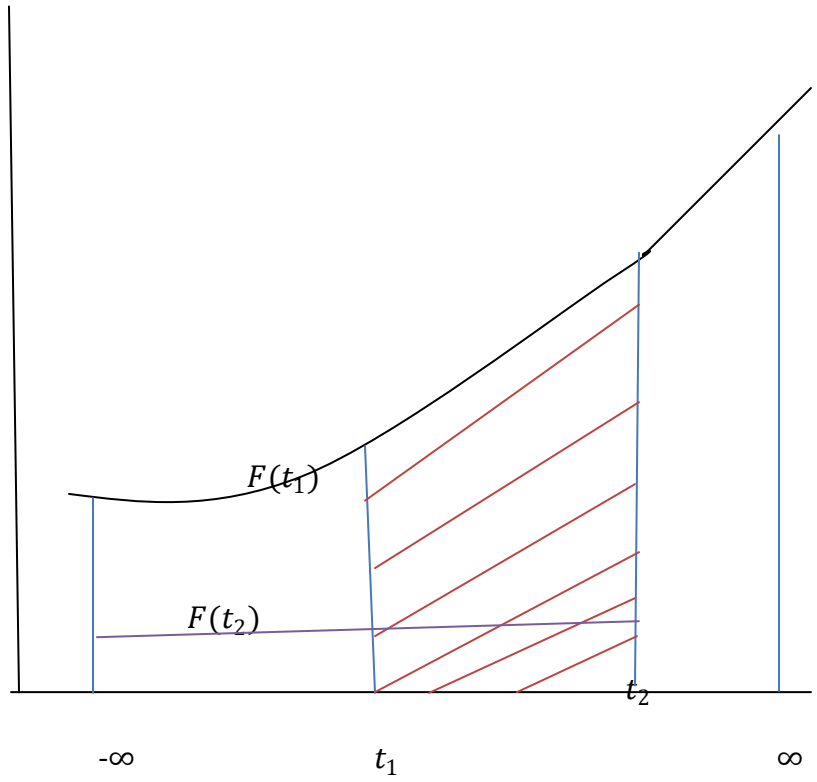
$Var(x) = 6.4 - \left(\frac{34}{15}\right)^2 = 1(\frac{59}{225})$

Cumulative distribution function (c.d.f)

For continuous random variables of x, the c.d.f at a particular point t in the range of the function is given by

$F(t) = P(x \leq t)$

$F(t) = \int_{-\infty}^{t} f(x)dx$



$P(t_1 < x < t_2) = F(t_2) - F(t_1)$

Example

A random variable x has the following p.d.f

$f(x) = \frac{3}{8}x^2 \quad 0 \leq x \leq 2$

Find c.d.f and sketch it

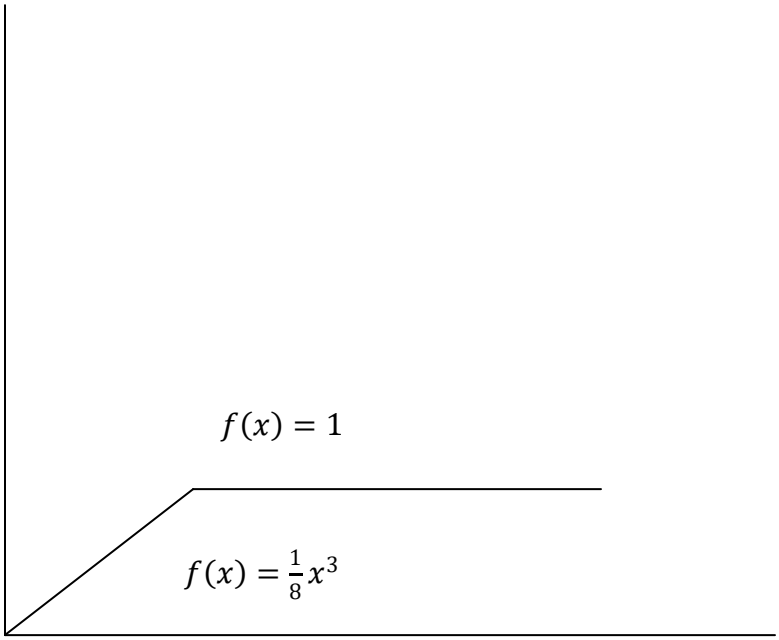
$P(X \leq t) = F(t)$

$= \int_0^t \frac{3}{8}x^2 \, dx$

$= \frac{3}{8} \times x^3 \times \frac{1}{3} = [\frac{x^3}{8}]_0^t$

$= \frac{t^3}{8}$

$$F(x) = P(X \leq x) = \begin{cases} 0 & x < 0 \\ \frac{1}{8}x^3 & 0 \leq x \leq 2 \\ 1 & x > 2 \end{cases}$$



Example

$$f(x) = \begin{cases} \frac{1}{4} & 0 \leq x \leq 2 \\ \frac{1}{4}(2x - 3) & 2 \leq x \leq 3 \end{cases}$$

Find c.d.f

$f(x)$ Is in two stages hence c.d.f $F(x)$ is in two stages

Stage 1

i) $0 \leq t \leq 2$

$F(t) = \int_0^t \frac{1}{4}\, dx$

$= \frac{1}{4}[x]_0^t$

$= \frac{1}{4}t$

$F(x) = \frac{1}{4}x$

$F(2) = \frac{1}{2}$ Maximum value

Stage 2

ii) $2 \leq t \leq 3$

$F(2) + \int_2^t \frac{1}{4}(2x - 3)dx$

$= \frac{1}{4}(\frac{2x^2}{2} - 3x)_2^t$

$= F(2) + \left(\frac{1}{4}t^2 - \frac{3}{4}t\right) - \left(1 - \frac{6}{4}\right)$

$= \frac{1}{2} + \frac{1}{4}t^2 - \frac{3}{4}t + \frac{1}{2}$

$F(x) = \frac{1}{4}x^2 - \frac{3}{4}x + 1$

$$F(x) = \begin{cases} 0 & x \leq 0 \\ \frac{1}{4x} & 0 \leq x \leq 2 \\ \frac{1}{4}x^2 - \frac{3}{4}x + 1 & 2 \leq x \leq 3 \\ 1 & x \geq 3 \end{cases}$$